BOARD OF INTERMEDIATE EDUCATION, KARACHI

INTERMEDIATE (11TH CLASS) EXAMINATION, 2024

CHEMISTRY

Past Paper 2024

Retyped faithfully from the original board paper — same questions, no changes.

Section 'A' (Multiple Choice Questions)

Maximum Marks: 17

Note: This section consists of 17 part questions and all are to be answered. Each question carries one mark. Do not copy the part questions in your answer book. Write only the answer in full against the proper number of the question and its part. The use of scientific calculator is allowed. (Options in the original are separated by star marks, shown here as A–D in printed order.)

Note: Time: 3 Hours — Maximum Marks: 85 (Science Pre-Engineering & Pre-Medical Groups)

Q1. Choose the correct answer for each from the given options:

Qi. Significant figures in 0.0880 are:

(A) 2 (B) 3

(C) 4 (D) 5

Qii. Alpha rays consist of:

(A) Two protons and two electrons (B) Two neutrons and two electrons

(C) Two protons and two neutrons (D) Two neutrons and one proton

Qiii. The number of orbitals in fourth energy level is:

(A) 4 (B) 9

(C) 16 (D) 32

Qiv. The number of sigma and pi bonds in C_2H_2 are:

(A) 2 and 3 (B) 2 and 2

(C) 3 and 2 (D) 3 and 3

Qv. This molecule has minimum bond angle:

(A) CS_2 (B) H_2O

(C) NH_3 (D) BF_3

Qvi. The geometry of NH_4^+ and SO_4^{2-} ions is:

(A) Tetrahedral
(B) Trigonal
(C) Pyramidal
(D) Square Planer

Qvii. These pairs do not obey Dalton's Law:

(A) He and H_2
(B) H_2 and Ar
(C) HCl and NH_3
(D) He and NH_3

Qviii. The vapour pressure of H_2O at 100°C is:

(A) 76 torr
(B) 5 atms
(C) 101325 Pascal
(D) 12.5 psi

Qix. London dispersion force is stronger in:

(A) F_2
(B) Cl_2
(C) Br_2
(D) I_2

Qx. Diamond is an example of:

(A) Metallic Solid
(B) Molecular Solid
(C) Ionic Solid
(D) Covalent Solid

Qxi. If $a \neq b \neq c$ and $\alpha = \gamma = 90°$ but $\beta \neq 90°$, then the crystal structure is:

(A) Orthorhombic
(B) Hexagonal
(C) Monoclinic
(D) Triclinic

Qxii. If a catalyst is added in a chemical system at equilibrium, the value of Kc:

(A) will decrease
(B) will increase
(C) will become zero
(D) will not be changed

Qxiii. This oxide is amphoteric:

(A) K_2O
(B) CO_2
(C) CaO
(D) Al_2O_3

Qxiv. If the rate law of reaction is R = K, its order of reaction is:

(A) Zero
(B) first
(C) second
(D) third

Qxv. The oxidation numbers of N in NH_4NO_3 are:

(A) -3, +5
(B) -3, +3
(C) -5, +5
(D) -1, +1

Qxvi. This compound gives acidic solution when dissolved in water:

(A) NaCl
(B) NH_4Cl
(C) CH_3COONH_4
(D) Na_2SO_4

Qxvii. At pH = 0, the universal indicator shows this colour:

(A) Red
(B) Orange
(C) Yellow
(D) Green

Section 'B' (Short-Answer Questions)

Maximum Marks: 36

Note: Answer any NINE part questions. All questions carry equal marks.

Q2.

(i) Define any four of the following: ☆ Stoichiometry ☆ Exponential notation ☆ Unit Cell ☆ Latent Heat of Fusion ☆ Energy of Activation

(ii) Chlorine gas is produced on an industrial scale by electrolysis of an aqueous solution of NaCl: $2NaCl + 2H_2O \rightarrow Cl_2 + H_2 + 2NaOH$. Calculate: ☆ The volume of Cl_2 at S.T.P produced from 58.5 g of NaCl. ☆ The total number of molecules of Cl_2 produced.

(iii) 53.5g of Aluminium is burnt with 22.4g of oxygen to given brilliant white flame of Al_2O_2. $4Al + 3O_2 \rightarrow 2Al_2O_3$. Which reactant is limiting reactant? Also find the mass of Al_2O_3 produced.

(iv) State 'Hund's rule'. Write electronic configuration of the following: ☆ Cr(Z = 24) ☆ Sr^{++}(Z = 38) ☆ S^{--}(Z = 16)

(v) Give any four scientific reasons for the following: ☆ Evaporation is a cooling process. ☆ Graphite conduct electricity but diamond does not. ☆ Boiling point of H_2O is higher than HF. ☆ Pressure cooker is used for rapid cooking. ☆ Milk sour more rapidly in summer than in winter.

(vi) A $100cm^3$ sample of hydrogen effuse four times as rapidly as $100cm^3$ of an unknown gas. Calculate the molecular mass of the unknown gas.

(vii) Drive Van der Waal's equation for real gases by correcting volume and pressure.

(viii) The solubility of $Mg(OH)_2$ at 25°C is 4.6×10^{-3}g/$100cm^3$, calculate the solubility product of the $Mg(OH)_2$.

(ix) Explain Bronsted Lowry Theory of acids and bases. Also explain conjugate acid-base pair.

(x) The reaction $2NO + Cl_2 \rightarrow 2NOCl$ was studied at 25°C the following results were obtained: Exp.1: [NO] = 0.1, $[Cl_2]$ = 0.1, Initial rate = 2.52×10^{-3} mol/dm^3.sec; Exp.2: [NO] = 0.1, $[Cl_2]$ = 0.2, Initial rate = 5.04×10^{-3} mol/dm^3.sec; Exp.3: [NO] = 0.2, $[Cl_2]$ = 0.1,

Initial rate = 10.05×10^{-3} mol/dm³.sec. On the basis of above information illustrate the Rate law and find the order of reaction.

(xi) Define Colligative properties. Explain elevation of boiling point and depression of freezing point.

(xii) Calculate the Heat of formation of Fe_2O_3 at 25°C: ☆ $2Fe + 3/2\ O_2 \rightarrow Fe_2O_3$, $\Delta Hf° = ?$ ☆ $H_2 + ½O_2 \rightarrow H_2O$, $\Delta H = -286$ KJ/mole ☆ $Fe_2O_3 + 3H_2O \rightarrow 2Fe(OH)_3$, $\Delta H = +289$ KJ/mole ☆ $Fe + 3H_2O \rightarrow Fe(OH)_2 + 3/2\ H_2$, $\Delta H = +161$ KJ/mole

(xiii) Differentiate any two of the following: ☆ VBT and MOT. ☆ Line Spectrum and Continuous Spectrum ☆ Ideal Solution and Non-ideal Solution

(xiv) At 30°C the vapour pressure of water is 31.6 torr. If 30g of urea (molecular weight = 60) are dissolved in 180g of water. Calculate the lowering of vapour pressure.

Section 'C' (Detailed Answer Questions)

Maximum Marks: 32

> Note: Attempt any Two questions from this section. All questions carry equal marks.

Q3.

(a) Write the postulates of Bohr's atomic model. Starting from $\Delta E = E_2 - E_1$ derive an expression for the wave number of radiations emitted by an electron.

(b) Define Corrosion. Balance the following equations by ion-electron method: ☆ $Cr(OH)_3 + SO_4^{2-} \rightarrow CrO_4^{2-} + SO_3^{2-}$ (Basic medium) ☆ $Cr_2O_7^{2-} + I_2 \rightarrow Cr^{3+} + IO_3^-$ (Acidic medium)

Q4.

(a) Write the postulates of VSEPR model. State the shapes of the following molecules with the help of hybridization: ☆ H_2O ☆ BF_3

(b) State law of mass action. Derive the expression of Kc for the general reversible reaction $aA + bB \rightleftharpoons cC + dD$ and also give the relationship between Kp, Kc and KA.

Q5.

(a) Define battery. Explain the working of lead storage battery and determined electrode potential of copper or zinc.

(b) Give the characteristics of ionic and covalent solids. Kc for reaction is 55 at 700°C. Calculate the equilibrium concentration of reactants and products when the concentration of each reactant is 2.55 mole/dm³. $H_2 + I_2 \rightleftharpoons 2HI$